

Lab 21 Worksheet — Cloud Service and Deployment Model Selection

Name: ____ Date: ____

Exam objective: Compare IaaS, PaaS and SaaS and the public, private, community and hybrid deployment models, and select the correct combination for a stated business requirement (Core 1 objective 4.1).

Record as you go

Step	What you did	What you observed
1	Build the shared-responsibility matrix with rows for networking, storage, servers, virtualization, operating system, middleware, runtime, data and applications.	
2	Mark for on-premises that the customer manages every layer, which is the baseline the three cloud models are measured against.	
3	Mark for IaaS that the provider manages up to virtualization and the customer manages the operating system and everything above it — examples are AWS EC2, Azure Virtual Machines and Google Compute Engine.	
4	Mark for PaaS that the provider manages up to the runtime and the customer manages only applications and data — examples are Azure App Service, Google App Engine and Heroku.	
5	Mark for SaaS that the provider manages every layer and the customer manages only their own data and users — examples are Microsoft 365, Google Workspace and Salesforce.	
6	Compare the four deployment models on cost, control, security, compliance and typical use: public, private, community and hybrid.	

Step	What you did	What you observed
7	Record the five cloud characteristics that define the model: shared resources, rapid elasticity, high availability, file synchronisation and metered utilisation.	
8	Explain metered utilisation and why it changes budgeting — you pay for what you consume, so an idle over-provisioned resource is pure waste.	
9	Solve the scenario of a hospital storing patient records under strict data residency rules, and justify why private cloud is correct and public is not.	
10	Solve the scenario of a startup needing email and document collaboration with no IT staff, and justify why SaaS on public cloud is correct.	
11	Solve the scenario of a retailer with steady baseline load and extreme seasonal peaks, and justify why hybrid cloud with cloud bursting is correct.	
12	Solve the remaining scenarios — several hospitals sharing a compliance platform, a development team needing full OS control, and a team deploying code without managing servers — and justify each.	

Verification

Success criterion: Your responsibility matrix correctly divides all nine layers across on-premises, IaaS, PaaS and SaaS; every scenario names both a service model and a deployment model; and each justification cites the specific requirement that ruled the alternatives out.

- I completed every step in the lab.
- My result meets the success criterion above.
- I recorded my evidence (screenshots, output, completed tables).

Reflection

What surprised you in this lab?

Where would you apply this on the job?

What do you still need to revise before the exam?
